Career Launcher MBA

QA Numbers

Review Test

M is the product of ten consecutive two-digit natural numbers. Y is the highest power of 2 in M. How many different values of M are possible such that the value of Y is maximum given?

(1) 1 (2) 2 (3) 3 (4) 4

Question – 1

N bells toll at different intervals which are multiples of two minutes. If all the bells toll together for the first time at 11:00 a.m. and 9th time at 3:00 a.m. on the next day, what is the maximum possible value of N?

(1) 60 (2) 24 (3) 16 (4) 12

Question – 2

A student wrote all the natural numbers from 2 to 10000 on a blackboard. Another student came and erased all the perfect squares. A second student came and erased all the perfect cubes. If students come this way and erase all the higher powers, find the number of students who erased at least one number?

(1) 6

(2) 7

(3) 12

(4) None of these

Question – 3

What is the unit digit of the following product:

$(8 – 5)(8^2 – 5^2)(8^3 – 5^3)\ldots\ldots (8^{2005} – 5^{2005})$

(1) 1 (2) 3

(3) 7 (4) 9

Question – 4

How many integers between 1 and 500 are divisible by exactly two of the three numbers 3, 5 and 7?

(1) 62 (2) 66

(3) 70 (4) 58

Question – 5

How many multiples of 8 from 10^6 to 10^{10} (both inclusive) are perfect squares?

(1) 24751

(2) 24750

(3) 12376

(4) 12375

Question – 6

For how many ordered pairs (x, y), where x and y are non-negative integers, is the equation $\sqrt{x} + \sqrt{y} = \sqrt{650}$ satisfied?

(1) 0 (2) 2 (3) 6 (4) 7

Question – 7

$2^{28} - 1$ is exactly divisible by two numbers between 120 and 130. The sum of these two numbers is:

(1) 255 (2) 256 (3) 253 (4) 257

Question – 8

You are selecting 5 numbers randomly out of the first 150 odd numbers. Sum of these 5 odd numbers is A. How many different values of A are possible?

(1) ${}^{150}C_5$

(2) 150

(3) 725

(4) 726

Question – 9

A box contains 100 tickets, numbered from1 to 100. A person picks out three tickets from the box, such that the product of the numbers on two of the tickets yields the number on the third ticket. How many tickets can never be picked up?

(1) 10 (2) 11 (3) 25 (4) 26

Question – 10

Find the last two digits of 26^{1979} ?

Question – 11

Find the remainder when 198^{79} is divided by 27?

Question – 12

If $x^{2a+1} + y^{2b}$ always ends in 0 and a, b are positive integers, then the values of x, y respectively are

(1) 21, 27

(2) 27, 29

(3) 21, 29

(4) 29, 21

Question – 13

Jaswant Singh had N number of chocolates with him. He started to distribute the chocolates among his kids according to ascending order of their age. First he gives x number of chocolates to the youngest kid, then x+ 5 chocolates to second youngest, x + 10 chocolates to third youngest and so on. After distributing among certain number of kids in the above manner he left with 7 chocolates and he realized that had he started with 15 more chocolates, he could have distributed chocolates to one more child in the same manner. Which of the following can be the value of N and x respectively?

(1) 34, 4

(2) 30, 9

(3) 26, 7

(4) All of the above

Question – 14

If the “n” th day of August is the same weeksday as the “2n” th day of October, then how many values of n are possible?

(1) 4 (2) 5 (3) 2 (4) 0

Question – 15

M is a natural number ($3 < M < 500$). How many values of M are possible such that HCF of M and 1250 is 1?

(1) 200

(2) 198

(3) 150

(4) 197

Question – 16

How many numbers below 100 are co-prime to 100 but not co-prime to 1500?

(1) 11 (2) 14 (3) 17 (4) 26

Question – 17

The micromanometer in a certain factory can measure the pressure inside the gas chamber from 1 unit to 999999 units. The problem with the instrument is that it always skips the digit 5 and hence moves directly from 4 to 6. What is the actual pressure inside the gas chamber if the micromanometer displays 003016?

(1) 2201

(2) 2202

(3) 2600

(4) 2960

Question – 18

The digits x, y and z of a three-digit natural number xyz satisfy the equation $121x + 11y + z = 567$. What is the sum of the digits of the three-digit natural number xyz?

(1)17

(2) 18

(3) 21

(4) Data Insufficient

Question – 19

In how many zeroes does $(10!)^{100!} \times (100!)^{10!}$ end in?

(1) 96

(2) $2^{24} + 24^{2}$

(3) $2^{100!} \times 24^{10!}$

(4) $2 \times (100!) + 24 \times (10!)$

Question – 20

There is an integer consisting of 2000 digits beginning with 3 as the leftmost digit. This integer is such that any two consecutive integers taken together is divisible either by 17 or 23. The last digit can be 'a' or 'b'. What is the value of 'a + b'?

(1) 4

(2) 5

(3) 7

(4) Cannot be determined

Question – 21

Take any 2-digit number and multiply the digits together. If this process is continued, all 2-digit numbers will become a single digit number. How many two digits will finish with zero?

(1) 13 (2) 20

(3) 24 (4) 19

Question – 22

Which factorial needs to be removed to make it a perfect square ?

1! × 2! × 3! × ……. × 119! × 120!

(1) 1!

(2) 59!

(3) 60!

(4) 120!

Question – 23

What is the sum of all positive integers less than 10^4 which are relatively prime to 10^4?

(1) 1.25×10^6

(2) 1.25×10^7

(3) 2×10^7

(4) 2×10^6

Question – 24

XX + YY + ZZ = XYZ where each of X, Y and Z represent distinct non zero digits, Z can be?

(1) 6

(2) 7

(3) 8

(4) 9

Question – 25

Thanks!